Ambiguity Analysis and Aliasing Mitigation for Wideband Sparse Extremely Large Aperture Arrays

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***Abstract*—Sparse distributed extremely large aperture arrays (ELAAs), formed by coherently combining spatially separated compact subarrays, are a foundational architecture for future 6G wireless systems, high-resolution radar sensing, and cooperative robotic perception. Their promise comes from meter-scale effective apertures synthesized without densely populating every antenna location. Their difficulty is equally fundamental: When the inter-subarray spacing greatly exceeds a half-wavelength, narrowband processing produces severe grating-lobe ambiguities, leading to high-amplitude ghost targets that can be indistinguishable from true targets. Wideband coherent processing can suppress these ghosts, but it also introduces a second ambiguity mechanism when implemented through discrete stepped frequencies. This paper develops a unified wideband ambiguity-function framework for sparse ELAAs under both far-field and near-field propagation models. We show that continuous wideband integration suppresses mismatched spatial hypotheses through sinc-weighted decorrelation induced by per-channel delay errors. In contrast, uniform stepped-frequency synthesis reduces to a Dirichlet kernel, creating periodic range-ambiguity replicas and revealing an inherent resolution–ambiguity tradeoff. To mitigate these replicas, we introduce randomized stepped-frequency processing, which preserves the continuous-frequency mainlobe in expectation while replacing deterministic ambiguity ridges with noise-like sidelobes whose fluctuation about the continuous-band kernel decays with the number of frequency steps. Numerical examples using a dual-module sparse ELAA demonstrate the transition from narrowband spatial aliasing to wideband sidelobe suppression, expose the structured range ambiguities caused by uniform stepping, and verify the effectiveness of randomized and optimized frequency hopping for ambiguity mitigation.**

***Index Terms*—Sparse distributed ELAAs, near-field sensing, 6G, integrated sensing and communications, coherent wideband processing, ambiguity function.**

I. Introduction

Sparse distributed extremely large aperture arrays (ELAAs), formed by coherently combining spatially separated compact subarrays, are emerging as a foundational architecture for future 6G wireless systems and high-resolution sensing platforms [1]–[3]. In next-generation cellular networks, distributed antenna panels separated by meters to tens of meters are expected to support spatial multiplexing, centimeter-level localization, and integrated sensing and communications (ISAC). The same principle appears in automotive radar with multi-module millimeter-wave arrays and in robotic perception with cooperative sensing nodes [4]–[7].

Across these applications, the central attraction of sparse ELAAs is clear: They synthesize a large effective aperture without the cost, power, and hardware complexity of densely populating the entire aperture. However, the same sparsity also creates a fundamental ambiguity problem. When the inter-subarray spacing D greatly exceeds a half-wavelength, narrowband processing produces severe grating lobes. In sensing applications, these grating lobes manifest as high-amplitude "ghost" targets that can be difficult to distinguish from true targets. In the far field, such ghosts appear as angular aliases; in the near field, spherical-wave propagation couples range and angle, so the ambiguity structures become curved and spatially varying rather than simple periodic angular replicas [8]–[10].

Wideband processing provides a powerful way to suppress these ambiguities. Intuitively, a false spatial hypothesis may align with the true target at one frequency but cannot remain phase-consistent across a large bandwidth, because the mismatch induces different propagation phase delays across the distributed aperture. Thus, bandwidth can convert spatial aliasing into frequency-dependent decorrelation. This phenomenon is important for sparse ELAAs, where narrowband aperture synthesis alone can be dominated by grating lobes. Yet the mechanism is not fully captured by conventional resolution arguments. Bandwidth does not merely improve range resolution; it reshapes the entire ambiguity surface of the array.

The practical implementation of wideband processing introduces an additional complication. Many radar and sensing systems already support large-bandwidth operation through wideband FMCW, MIMO, phased-array, and sub-THz radar hardware [11]–[15]. While stepped-frequency processing approximates continuous wideband integration, discrete frequency sampling can itself create periodic ambiguity replicas along the delay or range dimension. Therefore, sparse wideband ELAAs face a coupled design question: *how does bandwidth suppress spatial grating lobes, when does discrete frequency stepping create new range ambiguities, and how should frequency steps be selected to control this tradeoff?*

Existing work addresses parts of this question, but not

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their intersection. Near-field XL-MIMO and ELAA studies have examined spherical-wave propagation, range-dependent focusing, and large-aperture signal processing [2], [16], [17]. Recent analyses characterize localization limits and Cramér–Rao bounds for near-field or wideband systems [18], [19], while other works study ISAC tradeoffs for co-located MIMO channels or single-frequency near-field super-resolution [20], [21]. Complementary electromagnetic-domain signal-processing perspectives provide insights into continuous or connected apertures [22]. However, a unified ambiguity-function theory for *sparse, wideband, near-field* distributed arrays remains missing. In particular, existing analyses do not explain why wideband coherent processing suppresses the full-amplitude ghosts created by sparse spatial sampling, nor do they quantify how stepped-frequency implementations trade sidelobe suppression against range ambiguity.

This paper fills this gap by developing a wideband ambiguity-function (AF) framework for sparse ELAAs that reveals how frequency diversity reshapes sparse-aperture ambiguity. The main contributions are summarized as follows.

First, we formulate continuous- and discrete-frequency AFs for sparse ELAAs under both far-field and near-field propagation environments. This provides a common framework for comparing narrowband processing, ideal continuous wideband integration, uniform stepped-frequency synthesis, randomized frequency hopping, and optimized frequency-step design.

Second, we identify the mechanism of wideband grating-lobe suppression. We show that a mismatched spatial hypothesis produces per-channel delay errors, and that integration over bandwidth yields a sinc-weighted decorrelation factor. This result explains when and why bandwidth suppresses the ghost targets generated by sparse spatial sampling.

Third, we show that uniform stepped-frequency processing introduces a competing range-ambiguity mechanism. The frequency-synthesis kernel reduces to a Dirichlet kernel, producing periodic ambiguity replicas. This reveals a resolution–ambiguity tradeoff: Increasing the processed bandwidth improves range resolution and strengthens spatial decorrelation, but larger frequency steps reduce the unambiguous range.

Fourth, we introduce randomized stepped-frequency processing as a simple ambiguity-mitigation strategy. We show that randomized stepping preserves the ideal continuous-frequency mainlobe in expectation while spreading deterministic ambiguity ridges into noise-like sidelobes whose root-mean-square (RMS) fluctuation about the continuous-frequency kernel decreases as $\mathcal{O}(K^{-1/2})$, where K denotes the number of frequency steps. We further discuss optimized frequency-step design for reducing the peak sidelobe level beyond randomization alone.

Numerical examples based on a dual-module sparse ELAA validate the analysis. The results show that coherent wideband processing substantially suppresses the grating lobes of a sparse aperture, that uniform frequency stepping creates structured range replicas consistent with the Dirichlet-kernel prediction, and that randomized or optimized frequency hopping reduces these replicas. These findings suggest that ambiguity control, rather than nominal resolution alone, should be a primary design criterion for sparse wideband ELAAs in 6G ISAC, automotive radar, and cooperative robotic sensing.

II. System Model

A. Sparse ELAA Geometry

Consider a sparse ELAA consisting of m coherent subarrays, each a compact uniform linear array (ULA) of n elements with aperture L, aligned along the x-axis. The subarray centers are equally spaced by D:

$$x_i = \left(i - \tfrac{m+1}{2}\right) D, \quad i = 1, \ldots, m, \tag{1}$$

and the intra-subarray element offsets are $\delta x_p = -\frac{L}{2} + (p-1)\frac{L}{n-1}$, $p = 1, \ldots, n$. The position of the (p,i)-th sensor is thus $\boldsymbol{r}_{p,i} = [x_{p,i},\, 0]^{\mathsf{T}}$, where $x_{p,i} \triangleq x_i + \delta x_p$, and $(\cdot)^{\mathsf{T}}$ denotes transpose.

A target at polar coordinates (r, θ) has Cartesian position $\boldsymbol{p}(r,\theta) = [r\sin\theta,\ r\cos\theta]^{\mathsf{T}}$. Under spherical-wave propagation, the sensor-to-target distance is

$$d_{p,i}(r,\theta) = \sqrt{\left(r\sin\theta - x_{p,i}\right)^2 + \left(r\cos\theta\right)^2}. \tag{2}$$

B. Wideband Signal Model

Let $M = mn$ denote the total number of elements in the sparse ELAA. At frequency step f_k, $k = 1, \ldots, K$, the received array snapshot in the presence of Q targets is

$$\mathbf{y}(f_k) = \sum_{q=1}^{Q} S_q(f_k)\, \mathbf{a}(f_k; r_q, \theta_q) + \mathbf{w}(f_k), \tag{3}$$

where $S_q(f_k)$ denotes the reflection coefficient of the q-th target at frequency f_k, and $\mathbf{w}(f_k) \sim \mathcal{CN}(\mathbf{0}, \sigma^2\mathbf{I})$ is the additive noise vector. The entry of the steering vector corresponding to element p of subarray i is

$$\left[\mathbf{a}(f_k; r, \theta)\right]_{(i-1)n+p} = \exp\left(-j\frac{4\pi f_k}{c}\, d_{p,i}(r,\theta)\right), \tag{4}$$

where $d_{p,i}(r,\theta)$ is the spherical-wave propagation distance from the target at (r,θ) to the (p,i)-th element. The factor 4π reflects the round-trip path of an equivalent monostatic (phase-center) model, which is retained throughout; for a bistatic channel, $2d_{p,i}$ would be replaced by the sum of the transmitter-to-target and target-to-receiver distances.

This frequency-domain formulation is natural because each step frequency provides narrowband response with respect to the array aperture, while the wideband response is synthesized by coherent integration across $\{f_k\}_{k=1}^{K}$. Under near-field propagation the steering vector depends on range and angle, making (4) more appropriate than a far-field plane-wave model. The K frequency steps span a total bandwidth B around center frequency f_c. Throughout this paper, we assume narrowband operation at each step, phase coherence across subarrays and frequency steps, and a quasi-static scene over the coherent processing interval.

III. Wideband Ambiguity Function

Let η denote the parameter of interest for a target. In the far-field case, $\eta = \theta$, where θ is the DOA. In the near-field case, $\eta = \mathbf{p}$, where $\mathbf{p} = [x, y]^{\mathsf{T}}$ denotes the target position.

A. Definition

The continuous wideband AF between two targets characterized by parameters η_a and η_b is defined as the band-averaged normalized steering correlation [23]:

$$\mathcal{A}(\eta_a;\eta_b) \triangleq \frac{1}{B}\int_{f_c-\frac{B}{2}}^{f_c+\frac{B}{2}} \frac{\mathbf{a}(f;\eta_a)^{\mathrm{H}}\mathbf{a}(f;\eta_b)}{\|\mathbf{a}(f;\eta_a)\|_2\,\|\mathbf{a}(f;\eta_b)\|_2}\,df, \quad (5)$$

where $\|\cdot\|_2$ denotes the ℓ_2 norm. The discrete stepped-frequency AF is obtained by replacing the integral with a sum over the frequency steps $\{f_k\}_{k=1}^{K}$:

$$\mathcal{A}_S(\eta_a;\eta_b) \triangleq \frac{1}{K}\sum_{k=1}^{K} \frac{\mathbf{a}(f_k;\eta_a)^{\mathrm{H}}\mathbf{a}(f_k;\eta_b)}{\|\mathbf{a}(f_k;\eta_a)\|_2\,\|\mathbf{a}(f_k;\eta_b)\|_2}. \quad (6)$$

B. Far-Field Specialization

In the far field, the target parameter reduces to the DOA θ. We denote the far-field steering vector by $\mathbf{a}^{(\mathrm{FF})}(f;\theta)$ and the corresponding continuous- and step-frequency wideband AFs by $\mathcal{A}^{(\mathrm{FF})}(\theta_a;\theta_b)$ and $\mathcal{A}_S^{(\mathrm{FF})}(\theta_a;\theta_b)$.

Under plane-wave propagation, the steering entry for element p of subarray i is

$$\big[\mathbf{a}^{(\mathrm{FF})}(f;\theta)\big]_{(i-1)n+p} = \exp\!\left(j\frac{4\pi f}{c}\,x_{p,i}\sin\theta\right). \quad (7)$$

This follows from (4) with $d_{p,i} \approx r - x_{p,i}\sin\theta$ after removing the common range phase $e^{-j4\pi fr/c}$. Removing this term is legitimate for an angular slice in which both hypotheses share an equal (or compensated) reference range; a general wideband range–angle comparison must retain it. Since every entry has unit magnitude, $\|\mathbf{a}^{(\mathrm{FF})}(f;\theta)\|^2 = M$. Defining $\Delta u_{ab} \triangleq \sin\theta_b - \sin\theta_a$, the continuous far-field AF evaluates to

$$\mathcal{A}^{(\mathrm{FF})}(\theta_a;\theta_b) = \frac{1}{M}\sum_{i=1}^{m}\sum_{p=1}^{n}\exp\!\left(j\frac{4\pi f_c}{c}\,x_{p,i}\,\Delta u_{ab}\right) \times \operatorname{sinc}\!\left(\frac{2B}{c}\,x_{p,i}\,\Delta u_{ab}\right), \quad (8)$$

where $\operatorname{sinc}(x) = \sin(\pi x)/(\pi x)$, with $\operatorname{sinc}(0) = 1$. The first channel-wise zero occurs at $|x_{p,i}\,\Delta u_{ab}| = c/(2B)$, and beyond it the contribution of the (p,i)-th channel is bounded by the envelope $c/(2\pi B|x_{p,i}\,\Delta u_{ab}|)$.

C. Near-Field Specialization

In the near-field regime, the target parameter is the spatial position $\mathbf{p} = [r\sin\theta, r\cos\theta]^{\mathsf{T}}$. We therefore write the steering vector as $\mathbf{a}^{(\mathrm{NF})}(f;\mathbf{p})$ and the corresponding AFs as $\mathcal{A}^{(\mathrm{NF})}(\mathbf{p}_a;\mathbf{p}_b)$ and $\mathcal{A}_S^{(\mathrm{NF})}(\mathbf{p}_a;\mathbf{p}_b)$.

Using the spherical-wave propagation model in (2), the steering entry associated with element p of node i is

$$\big[\mathbf{a}^{(\mathrm{NF})}(f;\mathbf{p})\big]_{(i-1)n+p} = \exp\!\left(-j\frac{4\pi f}{c}\,d_{p,i}(\mathbf{p})\right), \quad (9)$$

where $d_{p,i}(\mathbf{p})$ denotes the propagation distance from the target at $\mathbf{p}$ to the (p,i)-th sensor. Unlike the far-field case, the phase now depends jointly on range and angle through the target position.

Since $\|\mathbf{a}^{(\mathrm{NF})}(f;\mathbf{p})\|^2 = M$, the continuous near-field AF is therefore

$$\begin{aligned}\mathcal{A}^{(\mathrm{NF})}(\mathbf{p}_a;\mathbf{p}_b) &= \frac{1}{MB}\int_{f_c-\frac{B}{2}}^{f_c+\frac{B}{2}} \mathbf{a}^{(\mathrm{NF})\mathrm{H}}(f;\mathbf{p}_a)\,\mathbf{a}^{(\mathrm{NF})}(f;\mathbf{p}_b)\,df \\ &= \frac{1}{M}\sum_{i=1}^{m}\sum_{p=1}^{n} e^{j\frac{4\pi f_c}{c}\delta d_{p,i}}\,\operatorname{sinc}\!\Big(\frac{2B}{c}\,\delta d_{p,i}\Big),\end{aligned} \quad (10)$$

where $\delta d_{p,i} \triangleq d_{p,i}(\mathbf{p}_a) - d_{p,i}(\mathbf{p}_b)$, and the discrete stepped-frequency version is

$$\mathcal{A}_S^{(\mathrm{NF})}(\mathbf{p}_a;\mathbf{p}_b) = \frac{1}{K}\sum_{k=1}^{K} \frac{(\mathbf{a}^{(\mathrm{NF})}(f_k;\mathbf{p}_a))^{\mathrm{H}}\mathbf{a}^{(\mathrm{NF})}(f_k;\mathbf{p}_b)}{\|\mathbf{a}^{(\mathrm{NF})}(f_k;\mathbf{p}_a)\|\,\|\mathbf{a}^{(\mathrm{NF})}(f_k;\mathbf{p}_b)\|}. \quad (11)$$

This formulation gives a joint range–angle description of the AF. In contrast to the far-field case, where the AF depends only on angular mismatch, the near-field ambiguity surface depends jointly on range and angle in the spatial domain, since it is determined by spherical-wave distance differences across the aperture.

IV. Wideband Sidelobe Suppression and Range Ambiguity

A. Coherent Wideband Focusing

For discrete-frequency signals, the coherent wideband matched response at hypothesis η is

$$z_{\mathrm{WB}}(\eta) = \frac{1}{K}\sum_{k=1}^{K}\frac{\mathbf{a}^{\mathrm{H}}(f_k;\eta)\mathbf{y}(f_k)}{\|\mathbf{a}(f_k;\eta)\|}. \quad (12)$$

Consider a single target at η_0 with frequency-flat reflection coefficient s, so that $\mathbf{y}(f_k) = s\,\mathbf{a}(f_k;\eta_0) + \mathbf{w}(f_k)$. Substituting into (12) and taking the expectation over noise yields

$$\mathbb{E}[z_{\mathrm{WB}}(\eta)] = \frac{s}{K}\sum_{k=1}^{K}\frac{\mathbf{a}^{\mathrm{H}}(f_k;\eta)\mathbf{a}(f_k;\eta_0)}{\|\mathbf{a}(f_k;\eta)\|}. \quad (13)$$

For the phase-only steering model adopted in this paper, every steering entry has unit magnitude, so $\|\mathbf{a}(f_k;\eta)\| = \sqrt{M}$ for all k and η. Defining the normalized response $\widetilde{z}_{\mathrm{WB}}(\eta) \triangleq z_{\mathrm{WB}}(\eta)/\sqrt{M}$, we obtain

$$\mathbb{E}[\widetilde{z}_{\mathrm{WB}}(\eta)] = s\,\mathcal{A}_S(\eta;\eta_0), \quad (14)$$

establishing that the expected wideband matched response is proportional to the discrete AF. The sidelobe behavior is therefore governed by $\mathcal{A}_S(\eta;\eta_0)$.

B. Sidelobe Suppression Mechanism

The key mechanism underlying sidelobe suppression is that a spatial mismatch is converted into a bandwidth-dependent delay mismatch. To make this explicit, define the per-channel round-trip delay mismatch in the sparse ELAA as

$$\Delta\tau_{p,i}(\eta,\eta_0) \triangleq \frac{2}{c}\Big(d_{p,i}(\eta) - d_{p,i}(\eta_0)\Big). \quad (15)$$

With this definition, the (p,i)-th entry of $\mathbf{a}(f;\eta)^{\mathrm{H}}\mathbf{a}(f;\eta_0)$ equals $e^{+j2\pi f\Delta\tau_{p,i}}$. In the continuous-frequency case, the

contribution of the (p,i)-th channel to the coherent wideband response is therefore

$$\frac{1}{B}\int_{f_c-\frac{B}{2}}^{f_c+\frac{B}{2}} e^{j2\pi f\,\Delta\tau_{p,i}}\,df = e^{j2\pi f_c\Delta\tau_{p,i}}\,\mathrm{sinc}\big(B\Delta\tau_{p,i}\big). \quad (16)$$

This shows that the response of a mismatched hypothesis is attenuated by a sinc factor whose width is controlled by the signal bandwidth B.

The finite-K response in (14) is a sample average over $\{f_k\}$. Replacing this average by continuous integration over the band, i.e., $\frac{1}{K}\sum_k(\cdot) \to \frac{1}{B}\int(\cdot)\,df$, yields the continuous-integration response $\bar{z}_{\mathrm{WB}}(\eta) \triangleq s\,\mathcal{A}(\eta;\eta_0)$. For the range-referenced far-field model, $\Delta\tau_{p,i} = \frac{2x_{p,i}}{c}(\sin\theta_0 - \sin\theta)$ and

$$\bar{z}_{\mathrm{WB}}(\theta) = \frac{s}{M}\sum_{i=1}^{m}\sum_{p=1}^{n} e^{j2\pi f_c\Delta\tau_{p,i}}\,\mathrm{sinc}\big(B\Delta\tau_{p,i}\big), \quad (17)$$

which coincides with (8) for $\theta_a = \theta$ and $\theta_b = \theta_0$. Since $|\mathrm{sinc}(B\Delta\tau)| \le \min\{1, 1/(\pi B|\Delta\tau|)\}$, each channel with a nonzero delay mismatch has its contribution bounded by an envelope that decays as $1/B$. Channels with $\Delta\tau_{p,i} = 0$ are not attenuated, and the sinc factor itself oscillates, so the attenuation is governed by a decaying envelope rather than being monotonic in B. This is the mechanism by which coherent wideband processing suppresses sidelobes and grating lobes in sparse ELAAs: a grating-lobe hypothesis whose carrier phases $2\pi f_c\Delta\tau_{p,i}$ align across the aperture still incurs large per-channel delay mismatches across the widely separated subarrays, and these contributions lose coherence over a sufficiently large bandwidth.

C. Range Ambiguity from Discrete Frequency Stepping

The sidelobe-suppression result above describes the benefit of wideband coherent integration. When the wideband response is synthesized from a *discrete* set of frequency steps, however, the same coherent integration also introduces a range-ambiguity mechanism. To see this, consider the frequency-synthesis kernel associated with an effective delay mismatch $\Delta\tau$:

$$S(\Delta\tau) \triangleq \frac{1}{K}\sum_{k=1}^{K} e^{j2\pi f_k\Delta\tau}. \quad (18)$$

When the frequencies are uniformly spaced as

$$f_k = f_c - \frac{B}{2} + (k-1)\Delta f, \qquad k = 1,\dots,K,$$

(18) reduces to

$$S_{\mathrm{unif}}(\Delta\tau) = e^{j\pi(2f_c - B + (K-1)\Delta f)\Delta\tau}\frac{\sin(\pi K\Delta f\,\Delta\tau)}{K\sin(\pi\Delta f\,\Delta\tau)}. \quad (19)$$

If the stepped frequencies exactly span the bandwidth B, so that $B = (K-1)\Delta f$, then (19) simplifies to

$$S_{\mathrm{unif}}(\Delta\tau) = e^{j2\pi f_c\Delta\tau}\frac{\sin(\pi K\Delta f\,\Delta\tau)}{K\sin(\pi\Delta f\,\Delta\tau)}. \quad (20)$$

At $\Delta f\,\Delta\tau = \ell \in \mathbb{Z}$, the singularities in (19) and (20) are removable: the Dirichlet ratio tends to $(-1)^{(K-1)\ell}$, so $|S_{\mathrm{unif}}| = 1$ at every delay $\Delta\tau = \ell/\Delta f$, i.e., the replicas have the full mainlobe magnitude. The first null occurs exactly at $\Delta\tau = 1/(K\Delta f)$.

This leads to a basic resolution–ambiguity tradeoff. When all channels share a common delay mismatch $\Delta\tau = 2\Delta R/c$, as for a pure range offset along a fixed direction, the first null corresponds to $\Delta R = c/(2K\Delta f) \approx c/(2B)$, and the replicas are spaced by $\Delta R_{\mathrm{amb}} = c/(2\Delta f)$. Increasing the total processed bandwidth B therefore improves range resolution and also strengthens wideband decorrelation across the aperture. However, when K is fixed, increasing B necessarily increases the frequency step size Δf. Consequently, ΔR_{amb} becomes smaller, and the unambiguous range shrinks accordingly. Therefore, uniformly stepped wideband processing may introduce structured ambiguity replicas, which motivates the use of randomized or otherwise nonuniform frequency stepping.

V. Randomized Frequency Stepping for Ambiguity Mitigation

A. Statistical Analysis of the RSF Kernel

Under randomized frequency stepping (RSF), let the frequency steps $\{f_k\}_{k=1}^{K}$ be independent and identically distributed (i.i.d.) samples drawn from $f_k \sim \mathcal{U}[f_c - \frac{B}{2},\, f_c + \frac{B}{2}]$, and define

$$S(\Delta\tau) \triangleq \frac{1}{K}\sum_{k=1}^{K} e^{j2\pi f_k\Delta\tau}.$$

Then $S(\Delta\tau)$ is the sample mean of i.i.d. unit-modulus complex random variables. Its mean and second moment are derived in Appendices A and B, respectively, and are given by

$$\mathbb{E}[S(\Delta\tau)] = e^{j2\pi f_c\Delta\tau}\,\mathrm{sinc}(B\Delta\tau), \quad (21)$$

$$\mathbb{E}\big[|S(\Delta\tau)|^2\big] = \frac{1}{K} + \Big(1 - \frac{1}{K}\Big)\big|\mathrm{sinc}(B\Delta\tau)\big|^2. \quad (22)$$

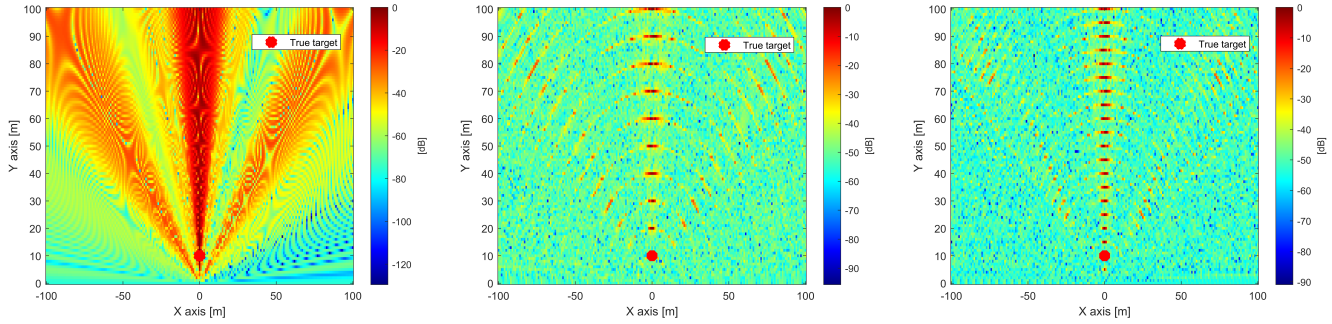
Equation (21) shows that RSF preserves the ideal continuous-frequency mainlobe in expectation. Writing $\mu(\Delta\tau) \triangleq e^{j2\pi f_c\Delta\tau}\mathrm{sinc}(B\Delta\tau)$, (21) and (22) give the exact centered variance

$$\mathbb{E}\big[|S(\Delta\tau) - \mu(\Delta\tau)|^2\big] = \frac{1 - |\mathrm{sinc}(B\Delta\tau)|^2}{K}.$$

Hence, the fluctuation of S about the continuous-band kernel has RMS order $K^{-1/2}$. The total RMS magnitude is approximately $K^{-1/2}$ only where $(K-1)|\mathrm{sinc}(B\Delta\tau)|^2 \ll 1$; at a fixed delay with $\mathrm{sinc}(B\Delta\tau) \neq 0$, it tends to $|\mathrm{sinc}(B\Delta\tau)|$, not to zero, as K increases. These pointwise moments do not bound the supremum over a search region, nor do they guarantee the level of any particular random or optimized realization.

Remark 1: Equation (22) consists of a finite-sample term $1/K$ and a coherent term governed by the continuous wideband kernel. Uniform stepping concentrates ambiguity energy into periodic full-magnitude peaks at $\Delta\tau = \ell/\Delta f$. Under RSF over the same band, $B = (K-1)\Delta f$ gives $\mathrm{sinc}(B\ell/\Delta f) = 0$ for $\ell \neq 0$, so $\mathbb{E}|S|^2 = 1/K$ at these delays: the deterministic ridges are replaced by noise-like sidelobes spread over the delay axis.

The spatial AF combines correlated channel kernels, because all channels share the same random frequencies. Let



(a) (b) (c)

Fig. 1. Near-field matched-response spectrum in the (x, y) plane for a target at $(0,\, 10\,\mathrm{m})$. (a) Narrowband. (b) Wideband, $\Delta f = 30\,\mathrm{MHz}$. (c) Wideband, $\Delta f = 60\,\mathrm{MHz}$.

$h(f) \triangleq \frac{1}{M}\sum_{i,p} e^{j2\pi f \Delta\tau_{p,i}}$, so that $\mathcal{A}_S(\eta;\eta_0) = \frac{1}{K}\sum_k h(f_k)$ and $\mathcal{A}(\eta;\eta_0) = \mathbb{E}_f[h(f)]$. Under i.i.d. uniform draws, $\mathbb{E}[\mathcal{A}_S] = \mathcal{A}$ and

$$\mathbb{E}\big[|\mathcal{A}_S - \mathcal{A}|^2\big] = \frac{\mathrm{Var}_f\, h(f)}{K} = \frac{\mathbb{E}_f|h(f)|^2 - |\mathcal{A}|^2}{K} \le \frac{1}{K},$$

where $\mathbb{E}_f|h(f)|^2 = \frac{1}{M^2}\sum_{\iota,\iota'} e^{j2\pi f_c(\Delta\tau_\iota - \Delta\tau_{\iota'})}\,\mathrm{sinc}\,\big(B(\Delta\tau_\iota - \Delta\tau_{\iota'})\big)$ and ι, ι' range over all channels (p, i).

B. Hopping Pattern Design

To further suppress ambiguity sidelobes, a fixed number K of distinct frequency steps can be selected from a discrete set of allowable bins $\mathcal{F} = \{\nu_\ell\}_{\ell=1}^{N_F} \subset [f_c - \frac{B}{2}, f_c + \frac{B}{2}]$ by minimizing the peak sidelobe level (PSL) of the delay kernel:

$$\min_{\{f_k\}_{k=1}^K \subset \mathcal{F}} \; \max_{\Delta\tau \in \mathcal{T}_{\mathrm{off}}} \left| \frac{1}{K}\sum_{k=1}^{K} e^{j2\pi f_k \Delta\tau} \right| \qquad (23)$$
$$\text{s.t. } |f_k - f_{k'}| \ge \Delta f_{\min}, \quad k \ne k'.$$

Here, $\mathcal{T}_{\mathrm{off}} = \{\Delta\tau : \tau_{\mathrm{ml}} \le |\Delta\tau| \le \tau_{\max}\}$ is a compact off-mainlobe delay region with a fixed mainlobe exclusion τ_{ml} and a maximum delay $\tau_{\max}$ set by the search region. The bound $\tau_{\max}$ is needed because, for bins on a uniform grid, the kernel is periodic in $\Delta\tau$ and attains unit magnitude at the inverse grid spacing, making an unbounded objective degenerate. The scalar objective is a surrogate for the PSL of the full spatial AF, which combines the channel delays $\Delta\tau_{p,i}$ as described above.

The problem in (23) can be handled using heuristic search methods such as genetic algorithms or particle swarm optimization. Two design insights follow from the kernel statistics. First, increasing K reduces the $1/K$ term in (22), so the fluctuation about the continuous-band kernel decreases as $\mathcal{O}(K^{-1/2})$, while the level approaches the continuous sinc kernel rather than zero. Second, flatter spectral coverage tends to preserve a narrow mainlobe and avoid concentrated coherent sidelobe peaks. These observations motivate optimized frequency-step selection as a way to shape the residual ambiguity structure beyond randomization alone.

VI. Numerical Examples

We consider a sparse ELAA with $m = 2$ subarrays, each a ULA of $n = 8$ elements with half-wavelength spacing at $f_c = 77\,\mathrm{GHz}$ ($\lambda_c \approx 3.9\,\mathrm{mm}$, subarray aperture $L \approx 1.4\,\mathrm{cm}$). The baseline separation is $D = 0.5\,\mathrm{m}$, yielding $D/(\lambda_c/2) \approx 256$ half-wavelength spacings between subarray centers.

A. Near-Field Two-Dimensional Spectrum

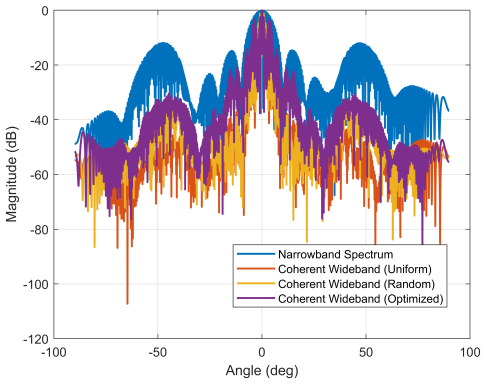
Fig. 1 displays the near-field 2D matched-response spectrum in the (x, y) plane for a single target located at $(0,\, 10\,\mathrm{m})$. The wideband processing uses $K = 100$ frequency steps. Narrowband focusing (Fig. 1(a)) produces strong sidelobes and ambiguity artifacts. Coherent wideband processing (Fig. 1(b)–(c)) yields a substantially cleaner localization map. As the step size increases from $\Delta f = 30\,\mathrm{MHz}$ to $60\,\mathrm{MHz}$, the decorrelation effect strengthens, but the periodic range-ambiguity replicas move closer together, consistent with the Dirichlet-kernel analysis in Section IV.

B. Comparison of Frequency-Step Configurations

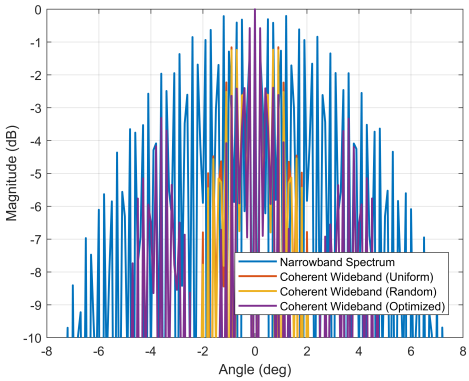
Although coherent wideband processing with uniform stepping suppresses the dominant grating lobes, the residual sidelobe level can still be significant. Fig. 2 compares the far-field angle spectrum under three frequency configurations: uniform stepping, i.i.d. uniform RSF, and a genetic-algorithm-optimized configuration. All three use $K = 100$ frequency steps over the same total bandwidth, $B = 10\,\mathrm{GHz}$, at an SNR of $15\,\mathrm{dB}$. The optimized configuration achieves the lowest PSL, $-2.39\,\mathrm{dB}$, compared with $-1.23\,\mathrm{dB}$ for the randomized configuration and $-1.16\,\mathrm{dB}$ for uniform stepping. These results show that frequency-step optimization can further reduce the PSL beyond randomization alone.

The same comparison is performed in the near field. For a target at $(0\,\mathrm{m}, 10\,\mathrm{m})$, Fig. 3 compares uniform stepping ($\Delta f = 300\,\mathrm{MHz}$, $K = 21$, $B = 6\,\mathrm{GHz}$), i.i.d. uniform RSF, and optimized stepping under the same total bandwidth. Uniform stepping produces structured range-ambiguity replicas, whereas RSF replaces these replicas with noise-like sidelobes. The optimized configuration provides a comparable but slightly lower PSL ($-8.60\,\mathrm{dB}$ versus $-8.53\,\mathrm{dB}$ for RSF).

Fig. 4 shows the average PSL over 100 Monte Carlo trials. RSF and optimized stepping consistently reduce the PSL relative to uniform stepping, confirming that nonuniform

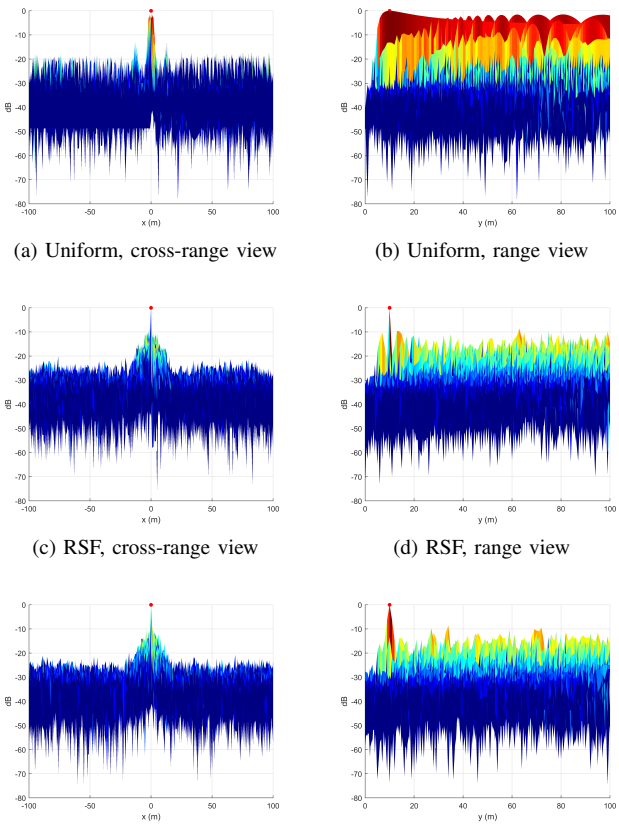


(a)



(b)

Fig. 2. Far-field angle spectrum under different frequency-step configurations. (a) Full view. (b) Mainlobe zoom.



(a) Uniform, cross-range view (b) Uniform, range view

(c) RSF, cross-range view (d) RSF, range view

(e) Optimized, cross-range view (f) Optimized, range view

Fig. 3. Near-field wideband processing under uniform stepping, RSF, and optimized frequency stepping. Left column: cross-range view; right column: range view.

frequency selection can mitigate the range replicas predicted by the Dirichlet-kernel analysis.

VII. Conclusions

We developed a unified wideband ambiguity-function framework for sparse ELAAs under both far-field and near-field propagation. The analysis shows that coherent wideband integration suppresses grating lobes through sinc-weighted decorrelation, while uniform stepped-frequency processing introduces periodic range ambiguities governed by a Dirichlet kernel. We further introduced randomized stepped-frequency processing, which preserves the continuous-band AF in expectation with an RMS fluctuation of order $\mathcal{O}(K^{-1/2})$, and showed numerically that randomized and optimized frequency selections reduce the observed PSL relative to uniform stepping. Future work will consider Doppler-inclusive ambiguity analysis and experimental validation.

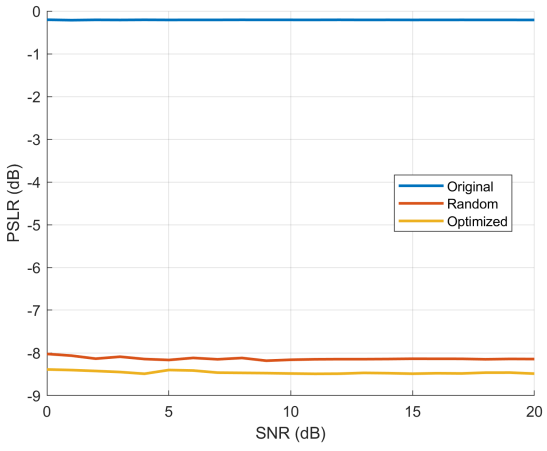


Fig. 4. Average PSL versus SNR for different frequency-step configurations.

Appendix

A. Derivation of $\mathbb{E}[S(\Delta\tau)]$

Let $X_k(\Delta\tau) \triangleq e^{j2\pi f_k \Delta\tau}$. Since $\{f_k\}_{k=1}^K$ are i.i.d., the random variables $\{X_k(\Delta\tau)\}_{k=1}^K$ are also i.i.d. Hence, by linearity of expectation and the uniform density $1/B$ on the

band,

$$\begin{aligned}\mathbb{E}[S(\Delta\tau)] &= \frac{1}{K}\sum_{k=1}^{K}\mathbb{E}[X_k(\Delta\tau)] = \mathbb{E}[X_1(\Delta\tau)] \\ &= \frac{1}{B}\int_{f_c-\frac{B}{2}}^{f_c+\frac{B}{2}} e^{j2\pi f\Delta\tau}\, df \\ &= e^{j2\pi f_c\Delta\tau}\,\frac{e^{j\pi B\Delta\tau}-e^{-j\pi B\Delta\tau}}{j2\pi B\Delta\tau} \\ &= e^{j2\pi f_c\Delta\tau}\,\mathrm{sinc}(B\Delta\tau),\end{aligned} \tag{24}$$

with $\mathrm{sinc}(0) = 1$ at $\Delta\tau = 0$. This is (21).

B. Derivation of $\mathbb{E}[|S(\Delta\tau)|^2]$

Expanding the squared magnitude gives

$$|S(\Delta\tau)|^2 = \frac{1}{K^2}\sum_{k=1}^{K}\sum_{\ell=1}^{K} X_k(\Delta\tau)\,\overline{X_\ell(\Delta\tau)}. \tag{25}$$

Taking expectation and separating diagonal ($k = \ell$) from off-diagonal ($k \neq \ell$) terms yields the following.

Diagonal terms (K in total): Since $|X_k(\Delta\tau)| = 1$ for every k,

$$\mathbb{E}\big[|X_k(\Delta\tau)|^2\big] = 1. \tag{26}$$

Off-diagonal terms ($K(K-1)$ in total): Since the frequency steps $\{f_k\}_{k=1}^{K}$ are i.i.d., the random variables $\{X_k(\Delta\tau)\}_{k=1}^{K}$ are also i.i.d. Therefore, for $k \neq \ell$, $X_k(\Delta\tau)$ and $\overline{X_\ell(\Delta\tau)}$ are independent, and

$$\begin{aligned}\mathbb{E}\Big[X_k(\Delta\tau)\,\overline{X_\ell(\Delta\tau)}\Big] &= \mathbb{E}[X_k(\Delta\tau)]\;\mathbb{E}\Big[\overline{X_\ell(\Delta\tau)}\Big] \\ &= |\mathbb{E}[X_1(\Delta\tau)]|^2 = \big|\mathrm{sinc}(B\,\Delta\tau)\big|^2.\end{aligned} \tag{27}$$

Combining the diagonal and off-diagonal contributions gives

$$\begin{aligned}\mathbb{E}\big[|S(\Delta\tau)|^2\big] &= \frac{K}{K^2} + \frac{K(K-1)}{K^2}\big|\mathrm{sinc}(B\,\Delta\tau)\big|^2 \\ &= \frac{1}{K} + \left(1-\frac{1}{K}\right)\big|\mathrm{sinc}(B\,\Delta\tau)\big|^2,\end{aligned} \tag{28}$$

which is the expression in (22).

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